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Roll No

EC-6002-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Signal Processing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define and explain discrete time linear time invariant system. 7
b) Discuss about the frequency domain representation of discrete time signals and system? 7
2. a) Describe the properties of Z-transform? 7
b) Explain analysis of linear time-invariant system in the Z-domain? 7
3. a) Define circular convolution. What is the difference between circular convolution and linear convolution? 7
b) Describe the properties of DFT. 7
4. a) How we can develop algorithm for 'N' composite number and explain? 7
b) What is decimation in frequency FFT algorithm? 7

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- 5. a) Write difference between FIR filter and IIR filter? 7
b) Explain the procedure for designing a FIR filter using the Rectangular window. 7
- 6. a) Discuss about inversion of the Z-transform. 7
b) Explain the difference between DFS and DFT. 7
- 7. a) What is bilinear transformation method of design IIR filter and explain? 7
b) Distinguish between stable and unstable system. 7
- 8. a) What is decimation in time FFT algorithm? 7
b) Discuss about the Kaiser window. 7
